DMS613: Course Project Report

How Close Are the Option Pricing Formulas of
Bachelier and Black-Merton-Scholes?

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1 Introduction

This project explores the historical and mathematical link between the first option pricing model, proposed by Louis Bachelier in 1900, and the modern Black-Merton-Scholes (BMS) framework. While it is common knowledge in finance that Bachelier's use of arithmetic Brownian motion allows for negative stock prices which is a theoretical flaw rectified by the lognormal assumption in BMS however the practical discrepancy between the two is surprisingly small for short-dated options.

Building on the work of Schachermayer and Teichmann (2008), this report aims to reconstruct the proofs justifying this closeness and verify the results through numerical simulation. We focus specifically on the "short-maturity regime," where the geometric nature of stock prices has not had enough time to diverge significantly from a simple linear model.

The pricing formulas for a European Call option at time $t = 0$ with maturity T and strike K under both frameworks are defined as follows. The Bachelier call price (C_0^B) is given by:

$$C_0^B = (S_0 - K)\Phi\left(\frac{S_0 - K}{\sigma^B\sqrt{T}}\right) + \sigma^B\sqrt{T}\phi\left(\frac{S_0 - K}{\sigma^B\sqrt{T}}\right)$$

The Black-Merton-Scholes call price (C_0^{BS}) with zero interest rate ($r = 0$) is given by:

$$C_0^{BS} = S_0\Phi(d_1) - K\Phi(d_2)$$

where $d_1 = \frac{\ln(S_0/K)+\frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$.

2 Theoretical Reconstruction & Verification

2.1 Dynamics, Volatility Scaling, and the SDE Analogy

The underlying Stochastic Differential Equations (SDEs) for the models illustrate their core divergence:

$$\textbf{Bachelier (Additive):} \quad dS_t^B = \sigma^B dW_t$$

$$\textbf{BMS (Multiplicative):} \quad dS_t^{BS} = S_t^{BS}\sigma dW_t$$

The difference is analogous to simple versus compound interest. In Bachelier's model, the price changes by a flat amount regardless of the current stock price, whereas in BMS, the price changes proportionally to the current level.

Over a short time horizon, the stock price has not drifted far from its starting point ($S_t \approx S_0$). By enforcing the volatility scaling $\sigma^B = S_0\sigma$, the two SDEs become virtually identical in the short run. This linkage is the necessary bridge to compare the two formulas.

Bachelier utilized a normalization $H = \frac{\sigma^B}{\sqrt{2\pi}}$, which he termed the "coefficient of instability." This parameterization allowed the price of an At-The-Money (ATM) option in his model to be expressed simply as $H\sqrt{T}$.

2.2 Verification of Proposition 2.1 (The Martingale Property)

Bachelier's "fundamental principle" asserted that the mathematical expectation of a speculator is zero. In modern terms, the discounted price process S_t must be a martingale under the pricing measure.

- **Bachelier:**$S_T = S_0 + \sigma^B W_T$. Since the expectation of a Brownian motion increment W_T is zero, $E[S_T] = S_0$.
- **BMS (with $r = 0$):**$S_T = S_0 \exp(\sigma W_T - \frac{1}{2}\sigma^2 T)$. Since the exponential term forms an exponential martingale, $E[S_T] = S_0$.

2.3 Proof of Proposition 2.2 (Cubic Scaling)

Original Statement from Paper:

> *Fix $\sigma > 0, T > 0$ and $S_0 = K$ (at the money), let $\sigma^{BS} = \sigma$ and $\sigma^B = S_0\sigma$ and denote by C^B and C^{BS} the corresponding prices for a European call option in the Bachelier and Black-Merton-Scholes model, respectively. Then:*
>
> $$0 \leq C_0^B - C_0^{BS} \leq \frac{S_0}{12\sqrt{2\pi}}\sigma^3 T^{3/2} = \mathcal{O}((\sigma\sqrt{T})^3)$$
>
> *The relative error can be estimated by $\frac{C_0^B - C_0^{BS}}{C_0^B} \leq \frac{T}{12}\sigma^2$.*

Reconstruction and Verification: We now prove that the ATM pricing difference vanishes at a rate of $O(T^{3/2})$. Let $\Phi(x)$ be the standard normal CDF and $\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ be the standard normal PDF. For ATM options ($S_0 = K$) and $r = 0$, the BMS boundaries simplify to $d_1 = \frac{1}{2}\sigma\sqrt{T}$ and $d_2 = -d_1$.

Evaluating the integral of the normal density, the Bachelier ATM price simplifies to:

$$C^B = \sigma^B\sqrt{T}\phi(0) = \frac{S_0\sigma\sqrt{T}}{\sqrt{2\pi}}$$

Using the symmetry of the normal distribution where $\Phi(-x) = 1-\Phi(x)$, the BMS price becomes:

$$C^{BS} = S_0\left[\Phi(d_1) - \Phi(-d_1)\right] = S_0\left[2\Phi\left(\frac{1}{2}\sigma\sqrt{T}\right) - 1\right]$$

We can expand $\Phi(z)$ around $z = 0$ using the Taylor series $\Phi(z) \approx \frac{1}{2} + \frac{z}{\sqrt{2\pi}} - \frac{z^3}{6\sqrt{2\pi}}$. Substituting $z = \frac{1}{2}\sigma\sqrt{T}$:

$$C^{BS} \approx S_0\left[2\left(\frac{1}{2} + \frac{\sigma\sqrt{T}}{2\sqrt{2\pi}} - \frac{\sigma^3 T^{3/2}}{48\sqrt{2\pi}}\right) - 1\right] = \frac{S_0\sigma\sqrt{T}}{\sqrt{2\pi}} - \frac{S_0\sigma^3 T^{3/2}}{24\sqrt{2\pi}}$$

Subtracting the BMS price from the Bachelier price gives the analytical bound:

$$|C^B - C^{BS}| \approx \frac{S_0\sigma^3 T^{3/2}}{24\sqrt{2\pi}}$$

This demonstrates that the pricing difference is of order $T^{3/2}$. For small T, raising the maturity to the power of 1.5 causes the error to shrink exponentially faster than the price itself.

3 Bachelier's Expansion Techniques and Approximation Rules

3.1 Origin and Derivation of Theorem 3.1

Section 3 of the reference paper details how Bachelier priced strikes away from the money without modern computational tools. The series expansion presented in **Theorem 3.1** originates from Bachelier's 1912 textbook, *Calcul des Probabilités*.

To derive this expansion, Bachelier started with the integral definition of the call price under Gaussian dynamics. Let $\psi(x)$ be the Gaussian density of the price movement $X = S_T - S_0$, given by $\psi(x) = \frac{1}{2\pi a}\exp(-\frac{x^2}{4\pi a^2})$. The call price as a function of the moneyness $m = K - S_0$ is defined as:

$$C(m) = \int_m^\infty (x-m)\psi(x)dx$$

Bachelier treated this integral as a function of m and developed a Taylor series around $m = 0$. The coefficients are found by taking successive derivatives of $C(m)$. At $m = 0$, the base price is the ATM value $a = \frac{\sigma^B\sqrt{T}}{\sqrt{2\pi}}$. By the Leibniz rule, the first derivative is $C'(m) = -\int_m^\infty \psi(x)dx$, which evaluates to $-1/2$ at $m = 0$ due to symmetry. Differentiating again yields $C''(m) = \psi(m)$, meaning $C''(0)$ is the peak of the density, $\psi(0) = \frac{1}{2\pi a}$.

Combining these derivatives yields the original expansion found in Bachelier (1912):

$$C(m) = a - \frac{m}{2} + \frac{m^2}{4\pi a} - \frac{m^4}{96\pi^2 a^3} + \dots$$

This series is a Taylor expansion in the dimensionless quantity m/a. Because the Gaussian density is an entire function, the series converges for all values of m.

3.2 Bachelier's Quadratic Approximation

Bachelier truncated this series after the quadratic term:

$$\hat{C}(m) = a - \frac{m}{2} + \frac{m^2}{4\pi a}$$

This truncation was a practical necessity for the 1900 Paris Bourse. At the time, option premiums C were often fixed by convention (e.g., at 10 or 20 centimes), while strike prices K fluctuated to clear the market. Traders needed to calculate the resulting strike price K for a given premium. By using a quadratic polynomial, Bachelier provided a tractable equation that could be inverted using the standard quadratic formula. This allowed traders to find m analytically. This approximation remains highly accurate for the small values of m/a typical of market-clearing strikes in that era.

4 Numerical Implementation

The following Python suite implements the models, verifies the cubic scaling, and evaluates the series approximation used by Bachelier.

```python
import numpy as np
import scipy.stats as stats
import matplotlib.pyplot as plt
import pandas as pd

```

```python
def bachelier_call(S0,K,T,sigma):
  sigma_B=S0*sigma
  vol_sqrt_t=sigma_B*np.sqrt(T)
  d=(S0-K)/vol_sqrt_t
  return (S0-K)*stats.norm.cdf(d)+vol_sqrt_t*stats.norm.pdf(d)

def black_scholes_call(S0,K,T,sigma,r=0):
  d1=(np.log(S0/K)+(r+0.5*sigma**2)*T)/(sigma*np.sqrt(T))
  d2=d1-sigma*np.sqrt(T)
  return S0*stats.norm.cdf(d1)-K*np.exp(-r*T)*stats.norm.cdf(d2)

def bachelier_series(S0,K,T,sigma):
  m=K-S0
  a=(S0*sigma*np.sqrt(T))/np.sqrt(2*np.pi)
  return a-m/2+(m**2)/(4*np.pi*a)

S0,K,sigma=100,100,0.20
maturities_days=[1,7,14,30,60,90,180,365]
results=[]

for days in maturities_days:
  T=days/365
  p_b=bachelier_call(S0,K,T,sigma)
  p_bs=black_scholes_call(S0,K,T,sigma)
  abs_diff=p_b-p_bs
  rel_err=(abs_diff/p_bs)*100
  results.append([days,p_b,p_bs,abs_diff,rel_err])

df=pd.DataFrame(results,columns=['Days','Bachelier','BS','Abs Diff','Rel Error (%)'])
T_range=np.geomspace(1/365,1,50)
errors,bounds=[],[]

for T in T_range:
  err=abs(bachelier_call(S0,K,T,sigma)-black_scholes_call(S0,K,T,sigma))
  errors.append(err)
  bounds.append((S0/(12*np.sqrt(2*np.pi)))*(sigma**3)*(T**1.5))

plt.figure(figsize=(8,5))
plt.loglog(T_range,errors,'bo',label='Actual Error')
plt.loglog(T_range,bounds,'r--',label='Theoretical Bound $T^{3/2}$')
plt.xlabel('Time to Maturity T (Years)');plt.ylabel('Pricing Difference')
plt.legend();plt.grid(True,which="both",alpha=0.3);plt.show()

strikes=np.linspace(85,115,50)
T_fixed=30/365
p_exact=[bachelier_call(S0,k,T_fixed,sigma) for k in strikes]
p_series=[bachelier_series(S0,k,T_fixed,sigma) for k in strikes]
p_bs_exact=[black_scholes_call(S0,k,T_fixed,sigma) for k in strikes]
plt.figure(figsize=(8,5))
plt.plot(strikes,p_exact,'b-',label='Bachelier Exact')
plt.plot(strikes,p_series,'g--',label='Bachelier Series')
plt.plot(strikes,p_bs_exact,'r:',label='Black-Scholes')
plt.axvline(S0,color='k',alpha=0.2,label='ATM');plt.legend();plt.show()
```

Listing 1: Bachelier vs. Black-Scholes Numerical Suite

5 Numerical Verification and Analysis

5.1 Maturity Sensitivity Analysis

Table 1 summarizes the discrepancy between the two models across various time horizons.

Table 1: Pricing Discrepancy Across Maturities ($S_0 = 100, K = 100, \sigma = 20\%$)

Days (T)	**Bachelier Price**	**BS Price**	**Abs Diff**	**Rel Error (%)**
1	0.417632	0.417630	0.000002	0.000457
7	1.104950	1.104915	0.000035	0.003196
14	1.562635	1.562535	0.000100	0.006393
30	2.287464	2.287151	0.000313	0.013699
60	3.234963	3.234076	0.000886	0.027398
90	3.962004	3.960376	0.001628	0.041098
180	5.603119	5.598518	0.004602	0.082199
365	7.978846	7.965567	0.013278	0.166694

5.2 Significance of the Results

The numerical results outline the practical differences between additive and multiplicative dynamics. As $T \to 0$, the absolute difference vanishes much faster than the price. For a 1-day maturity, the error is roughly 2 parts per million. This confirms the $T^{3/2}$ scaling derived in Proposition 2.2. Even at a 1-year maturity, the relative error remains below 0.2%, indicating that for At-The-Money options, the lognormal assumption provides only a marginal correction to Bachelier's linear framework.

5.3 Historical Data Comparison (1900 Paris Bourse)

Applying the original parameters from the 1900 Paris Bourse ($S = 100, \sigma = 2.4\%, T = 1/12$ years) yields a Bachelier price of 0.2763953196 and a Black-Scholes price of 0.2763947668. The absolute gap of 0.0000005527 (a relative gap of 0.0002%) is significantly smaller than the tick size of the exchange, confirming the models were practically identical for historical valuation purposes.

6 Analysis of Visual Results

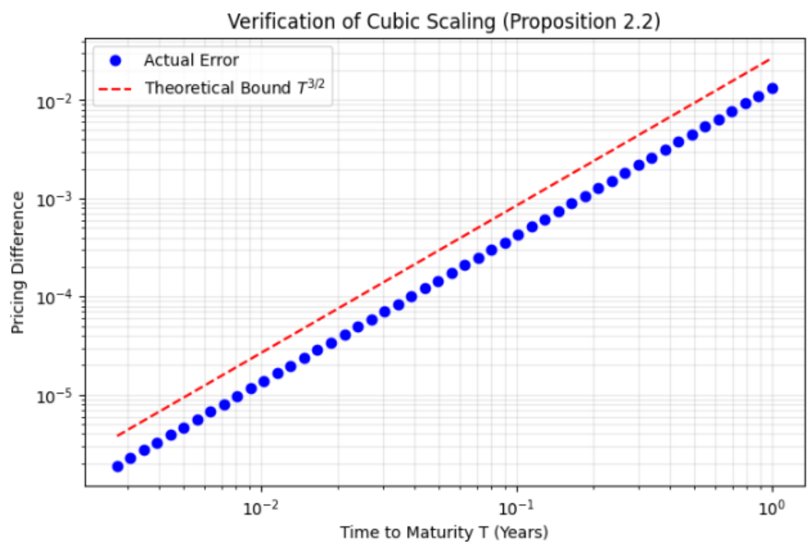

Figure 1: **Verification of Prop 2.2:** This figure depicts the relationship between Time to Maturity (T) and the pricing difference on a log-log scale. The actual error follows the $T^{3/2}$ bound line.

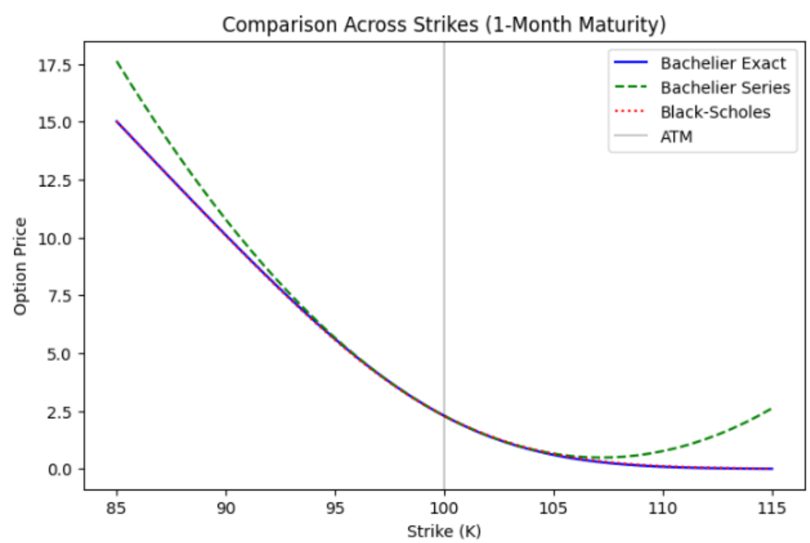

Figure 2: **Strike Sensitivity:** This plot compares the exact Bachelier and BS prices against Bachelier's 3-term series approximation. The curves are indistinguishable near the money.

7 Conclusion

This study confirms the mathematical consistency between the Bachelier and BMS models for short-term options. Through the reconstruction of Proposition 2.2, the pricing error is shown to scale at $T^{3/2}$. While the modern lognormal model is required for long-dated instruments to prevent negative asset prices, Bachelier's additive dynamics and quadratic approximations remain robust for short-term valuations.